

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
“JNANA SANGAMA”, BELAGAVI-590018



Project report on
“DESIGN AND DEVELOPMENT OF A DECENTRALIZED SWARM ROBOTICS
WAREHOUSE SYSTEM”

BACHELOR OF ENGINEERING
DEPARTMENT OF MECHATRONICS

Submitted by:

S. AKSHAY [10X23MT023]

SHASHANK M. [10X23MT025]

SUBASH G. [10X23MT027]

N. SUDHARSANA LAXMI [10X23MT028]

under the guidance of
Mr. Sandeep D. Dixit
Associate professor Dept. of Mechatronics, TOCE, Bengaluru



Department of Mechatronics
THE OXFORD COLLEGE OF ENGINEERING
Bengaluru-560068
2025-2026

THE OXFORD COLLEGE OF ENGINEERING

BENGALURU-560068

Department of Mechatronics



CERTIFICATE

Certified that the Project work entitled “**DESIGN AND DEVELOPMENT OF A DECENTRALIZED SWARM ROBOTICS WAREHOUSE SYSTEM**” is a bonafide work carried out by, **S. AKSHAY (10X23MT023), SHASHANK M. (10X23MT025), SUBASH G. (10X23MT027), N. SUDHARSANA LAXMI (10X23MT028)**. It is certified that all the corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the Department Library. The Project report has been approval as it satisfies the academic requirements in respect of project work prescribed for the said degree.

Signature of Guide

Mr. Sandeep D. Dixit

Signature of HOD

Dr. Madhura S.

THE OXFORD COLLEGE OF ENGINEERING

BENGALURU-560068

Department of Mechatronics



DECLARATION

We, **S. AKSHAY (10X23MT023)**, **SHASHANK M. (10X23MT025)**, **SUBASH G. (10X23MT027)**, and **N. SUDHARSANA LAXMI (10X23MT028)** students of semesters VI B.E. Mechatronics Engineering, The Oxford College of Engineering, hereby declare that the project work entitled “**DESIGN AND DEVELOPMENT OF A DECENTRALIZED SWARM ROBOTICS WAREHOUSE SYSTEM**” has been independently carried out by us under the guidance of **SANDEEP D DIXIT**, Professor, Department of Mechatronics, The Oxford College of Engineering, Bangalore.

We declare that, to the best of our knowledge and belief, the work reported here is not part of any other report based on which a degree or award was conferred on an earlier occasion on this by any other student.

PLACE: BANGALORE

DATE :

S. AKSHAY	[10X23MT023]
SHASHANK M.	[10X23MT025]
SUBASH G.	[10X23MT027]
N. SUDHARSANA LAXMI	[10X23MT028]

ACKNOWLEDGEMENT

The satisfaction and euphoria that accompany the successful completion of any task would be incomplete without the mention of the people who made it possible with continuous guidance and encouragement and crowned our efforts with success.

We have great pleasure in expressing our deep sense of gratitude to Late **Sri. S Narasa Raju**, Founder Chairman and we consider ourselves proud to be a part of the Oxford family, the institution that stood by our way in all our endeavours. So, we express our humble gratitude to **Sri S N V L Narasimha Raju**, Chairman, The Oxford College of Engineering for providing all facilities for our work to be a better one.

We would like to express our gratitude to **Dr. Rakesh S.G.**, Principal, The Oxford College of Engineering for providing congenial environment surroundings to work in. Words are inadequate in offering our thanks to **Dr. Madhura S.**, Head of Department of MTE and our guide **Mr. Sandeep D. Dixit**, Asst. Professor of MTE, for the moral inspiration, encouragement, and valuable guidance throughout the course of this work. We also thank all the staff members of the Mechatronics department and all those who have helped us with their valuable suggestions in the successful completion of this Project Work.

We also express our heartfelt thanks to our parents for being there during success and during turbulent times. As a dedication, we would dedicate this effort to the one who is omnipotent and omnipresent for having blessed us as generously as ever “The Almighty”.

S. AKSHAY	[10X23MT023]
SHASHANK M.	[10X23MT025]
SUBASH G.	[10X23MT027]
N. SUDHARSANA LAXMI	[10X23MT028]

TABLE OF CONTENTS

SL. No	Title	Page No.
i	Abstract	01
01.	Chapter 1	02
1.1	Introduction & Project Architecture	02
1.2	Decentralized Swarm Principles	02
1.3	Structural Data Flow Loop	02
02.	Chapter 2	03-09
2.1	Working Principle	03
2.2	Algorithms / Techniques	03
2.3	ArUco Marker Detection and Pose Estimation	03
2.4	Perspective Projection and Homography	04
2.41	Direct Linear Transform (DLT)	04
2.4.2	Differential Drive Kinematics	05
2.4.3	Principle of Virtual Work & Scissor Linkage Mechanics	05
2.5	Data Collection / Data Acquisition	06
2.6	Python Digital Twin & Tracking Dashboard	06
2.7	Homography & Real-to-Pixel Coordinate Scaling	06-07
2.8	Comprehensive Core Dashboard Engine Script	07
03.	Chapter 3	10-11
3.1	Mechanical Design & Structural Body Modeling	10
3.2	Tiered Chassis Topology	10
3.3	Material Selection & Manufacturing Feasibility	11
3.4	Mechanical Gripper Linkage Geometry	11
04.	Chapter 4	12-13
4.1	Electrical Architecture & Hardware Integration	12
4.2	Core Pin Connection Profile	12
4.3	Motor Driver Power Split Rules	13

05.	Chapter 5	14-19
5.1	System Development	14
5.2	Module Description	14
5.3	Integration of Components	14-15
5.4	Fabrication / Assembly Process	15
5.5	Onboard Decentralized Swarm Agent Firmware	15
5.6	Onboard Embedded Agent C++ Engine Code	15-17
5.7	System Verification & Presentation Benchmarks	17
5.8	UDP Broadcast Architecture in ESP32 Networks	18
5.9	Packet Loss Benchmarks, Latency Optimization & Power Save Mitigation	18-19
5.10	Onboard Embedded Agent Firmware & Local Edge Computation	19
06.	Chapter 6	20-21
6.1	Proportional Trajectory Error Realignment	20
6.2	Tactile Sensor Fusion Handoff	20-21
07.	Chapter 7	21
7.1	References	21-22

TABLE OF FIGURES

SL. No	Title	Page No.
1.1	Online ArUco Tracking Token Matrices (ID 0 and ID 1)	07
1.2	Live Visual Representation of the Real-Time Metric Swarm Dashboard	09
1.3	3D CAD Geometric Structural Shell Layout – Lower Deck Profile (LD)	10
1.4	3D CAD Geometric Structural Shell Layout – Upper Deck Profile (UD)	10
1.5	Physical Prototyping Assembly & End-Effector Scissor Gripper Mechanism Close-ups	11
1.6	Schematic Block Diagram of the Whole Integrated Circuit Distribution System	13
1.7	Onboard Embedded Agent C++ Engine Code	17
1.8	System Verification, Presentation Benchmarks	21

TABLE OF TABLES

SL. No	Title	Page No.
1.1	Core Pin Connection Profile	12